02/04/2022

Monday, January 31, 2022 12:07 PM

M168_Notes_3.1

Setting Up Linear Programming Problems

Example

A company produces handmade skillets in two sizes, big and giant. To produce one big skillet requires 3 lbs of iron and 6 minutes of labor. To produce one giant skillet requires 4 lbs of iron and 3 minutes of labor. The profit for each big skillet is \$20 and the profit for each giant skillet is \$25. If 1000 lbs of iron and 20 hours of labor are available each day, how many each size skillet should be made to maximize profits?

$ax \; by + c \geq 0$

x = # big skillets

y = # giant "

Objective: $P = 20x + 25y$

Subject to

$3x + 4y \leq 1000$ (Amount of iron in lbs)

$6x + 3y \leq 1200$ (Amount of labour)

$x \geq 0, \; y \geq 0$

Example

Annika has \$3000 available to invest, with which she buys shares of stock in the Bass Company and Forte Inc. A share of Bass stock costs \$10 and pays dividends of \$3.00 per share each year. Forte stock costs \$50 per share and pays dividends of \$4.00 per share each year. Annika wishes to purchase no more than 80 total shares of stock in order to avoid paying higher trading fees to her broker. Due to the volatility of the Bass stock, her broker recommends that she invest in at most 50 shares of that company's stock. Determine the number of shares Annika should buy of each kind of stock in order to maximize the amount from dividends that she will receive at the end of the first year.

x = # shares of Bass Comp.

y = # " " Forte Inc.

Obj. $D = 3x + 4y$

Subject to:

$$x + y \leq 80$$
$$x \leq 50$$
$$10 \cdot x + 50y \leq 3000$$
$$x \geq 0, \quad y \geq 0$$

Example

A dietitian is to prepare two foods in order to meet certain requirements. Each ounce of food I contains 100 units of vitamin C, 40 units of vitamin D and 20 units of vitamin E and costs 25 cents. Each ounce of food II contains 10 units of vitamin C, 80 units of vitamin D and 15 units of vitamin E and costs 15 cents. The mixture of the two foods is to contain at least 260 units of vitamin C, 320 units of vitamin D and 120 units of vitamin E. How many ounces of each type of food should be used in order to minimize the cost?

Example

A craftsman has 150 units of wood, 90 units of glue and 150 units of paint. A small picture frame requires 1 unit of wood, 1 unit of glue and 2 units of paint while a large picture frame requires 5, 2 and 1 respectively. If a small frame sells for $175 and a large frame for $400, how many of each should be made to maximize the revenue?

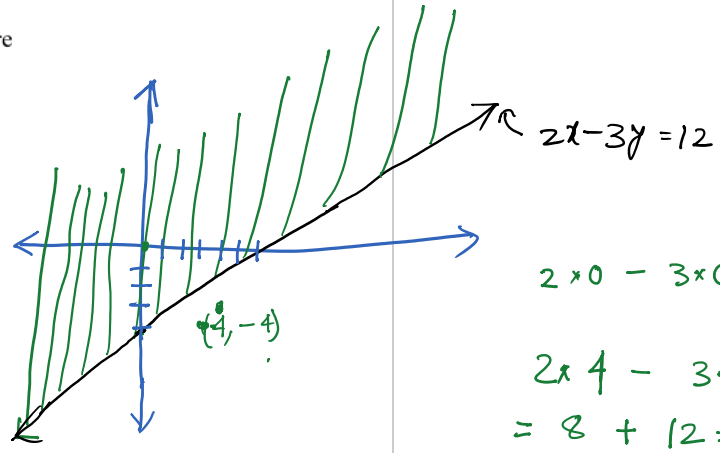
Graphing Systems of Linear Inequalities

The general forms for linear inequalities are

$ax + by + c \geq 0$
$ax + by + c > 0$
$ax + by + c \leq 0$
$ax + by + c < 0$

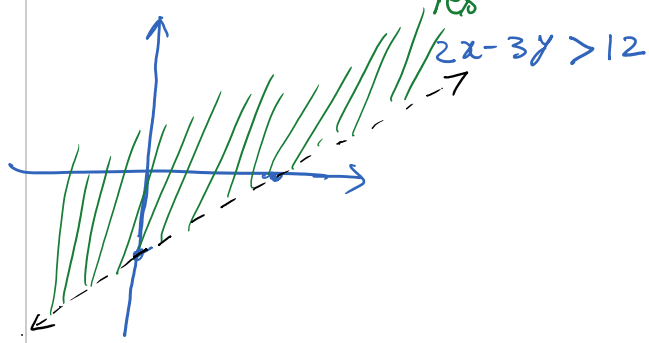
Example
Graph $2x - 3y \geq 12 \Leftrightarrow 2x - 3y - 12 \geq 0$

$(0, -4)$
$(6, 0)$



$2 \times 0 - 3 \times 0 = 0$

$2 \times 4 - 3 \times (-4)$
$= 8 + 12 = 20 \overset{?}{\geq} 12$
Yes



NOTE - if your line passes through the origin, you must take a different point for a test point.
If our inequality had ≥ or ≤ we draw the bounding line as a solid.

If our inequality had > or < we draw the bounding line as DASHED .

The region that satisfies our inequality is called the *feasible region*. This is the region that is white (unshaded). Please label it with an *S*.

What if we have two inequalities (a system)? The feasible region (solution) will be where they are both true at the same time.

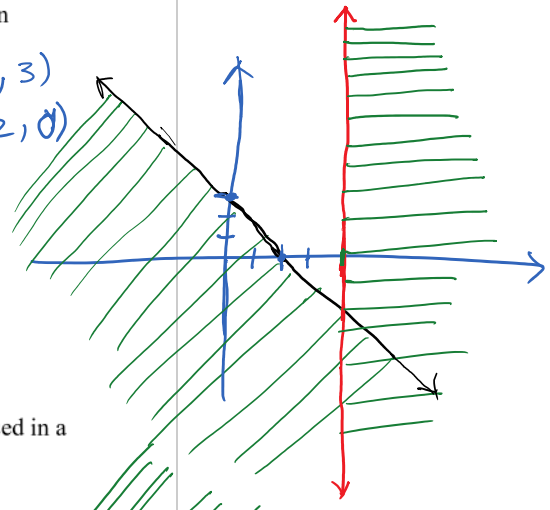
Example

Find the feasible region for the system $3x + 2y \geq 6$

$\cancel{x \leq 4}$ $x \leq 4$

(0, 3)
(2, 0)

$3 \times 0 + 2 \times 0 \overset{?}{\geq} 6$
NO

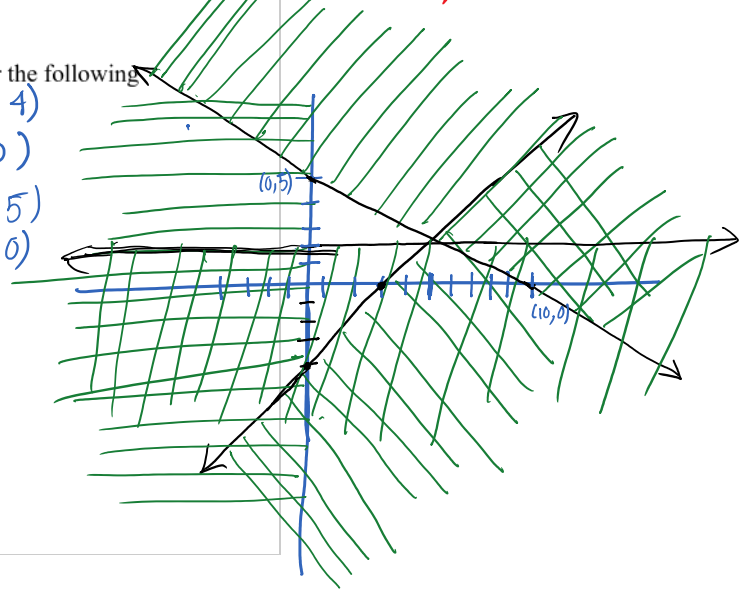


It is *UNBOUNDED* because the feasible region cannot be enclosed in a circle.
If S can be enclosed by a circle, it is called *BOUNDED*.

Example
Find the feasible region and label the corner points for the following system of linear inequalities:

$\cancel{4x - 3y \leq 12}$ $4x - 3y \leq 12$ (0, −4) (3, 0)

$\cancel{x + 2y \leq 10}$ $x + 2y \leq 10$ (0, 5) (10, 0)

$\cancel{x \geq 0}$ $x \geq 0$

$\cancel{y \geq 2}$ $y \geq 2$